Why are arrays useful? What is broadcasting? Why do we need to know this? How are arrays in numpy different from those in Xarray?

Numbers in python or in most coding languages are stored in computer memory before being divided into arrays so it is useful to understand the underlying conceptual model of how computes store numbers in memory. This memory can be visualized as a long sequence of 0s and 1s and something being stored in memory simply means that they are encoded as a series of binary values that can be changed to present different information. Meaning that storing something in memory is recording it as a specific pattern of 0s and 1s that correspond to a specific value.

Numpy stores vectors as a structured row of values that correspond to each other and anything higher than a 2D array can be seen as being composed of 2D arrays. An index is a value that corresponds to the order of the values of a vector or array. Multidimensional vectors contain axes and rows that are necessary for understanding the arrays. Reductions of an array are used to change the order of the array.

All operations then completed in the array are done element by element and would usually require the array to have the same chape to have operators used on them. However, numpy has concepts called broadcasting and conformant arrays that will take the current array and make sure it is the exact same shape and rank with other axes.

So, arrays are useful as they store multidimensional data compactly and broadcasting is numpy's rule for automatically stretching and bending arrays so that arrays of different shapes can be combined. Xarray differs from number as it adds named dimensions, coordinates, and other metadata that can solidified variables such as time or latitude other than arbitray numbers.